

Design of Pascal band-pass filter for EMG signals

The interface is opened and the design specifications are entered.

Design specifications

Ripple [dB]: Attenuation [dB]: ω_s/ω_p :

Calculate filter order

☐ Butterworth

☐ Chebyshev I

☐ Chebyshev...

☐ Elíptico

☒ Pascal

Bessel Thomson

Transfer function Bode of the approximations Finish

Pascal is selected and the transfer function of the real case is calculated.

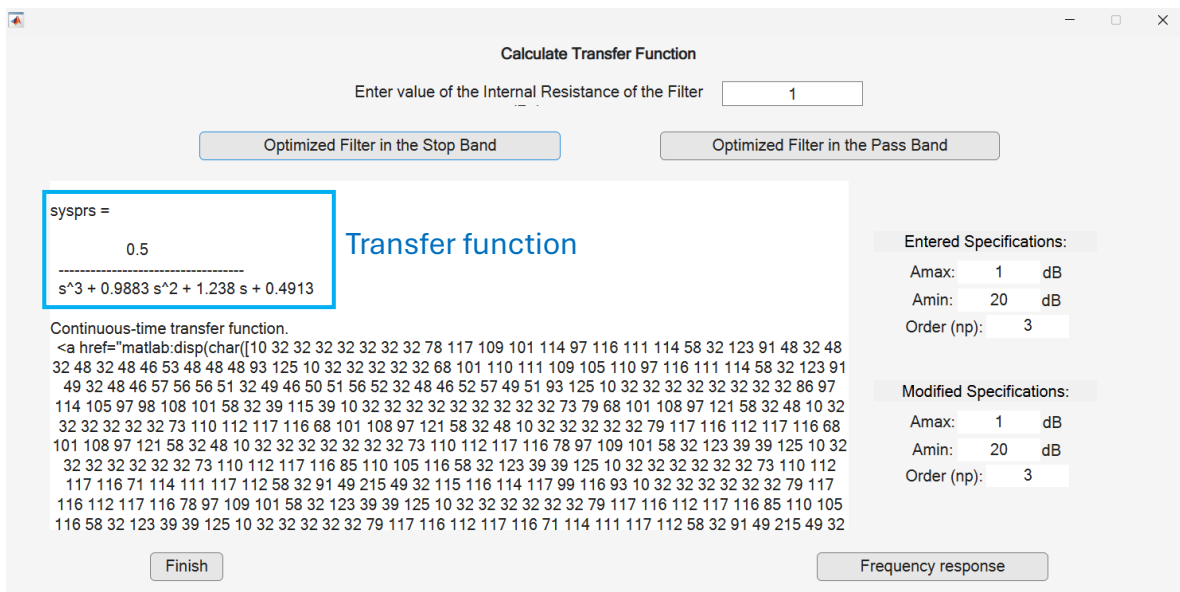
Determine Ideal Transfer Function of the Filter

Determine the actual Transfer Function of the Filter.

Choose another approach

Finish

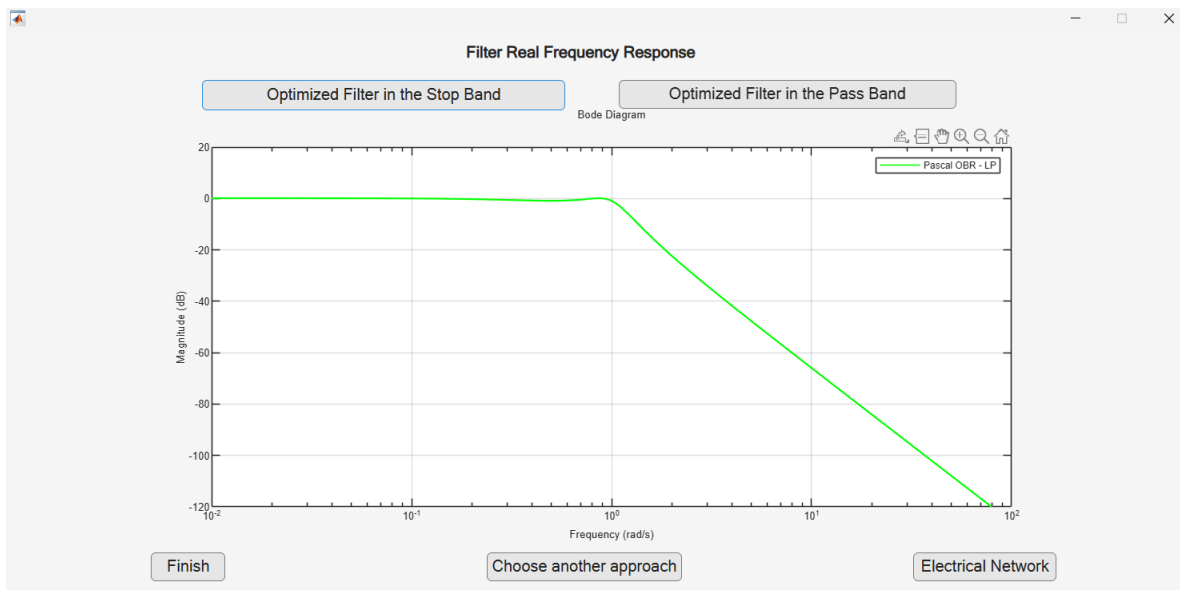
The option optimized filter in the stop band is selected.



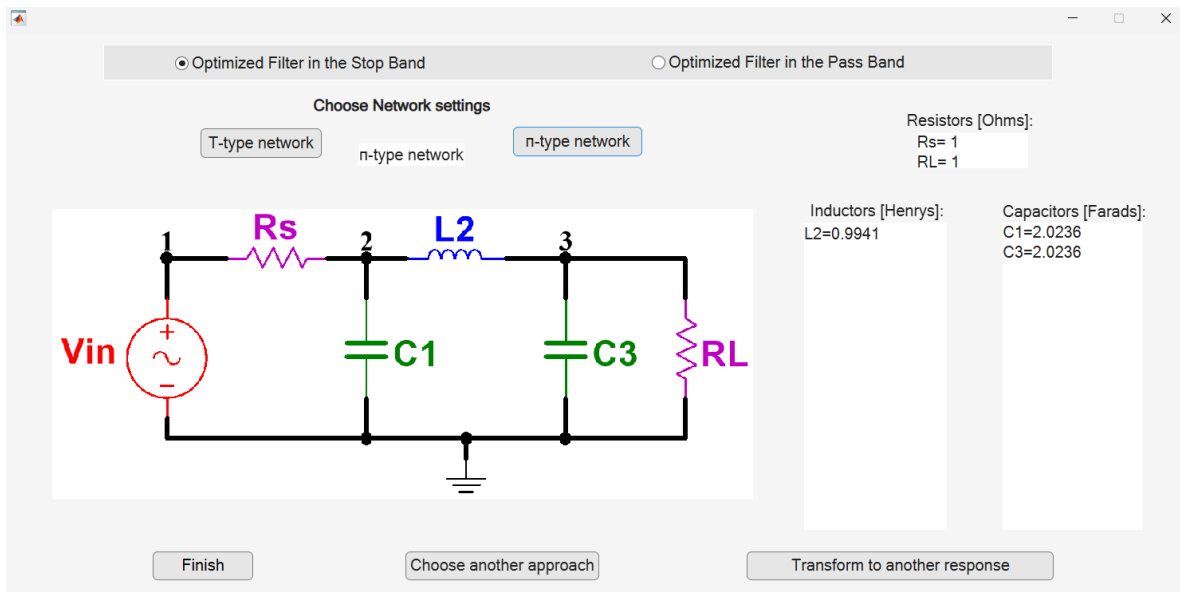
The way of printing the transfer function in the most current versions of MATLAB numerically drags internal structures of the display.

The transfer function obtained is the one registered in Table III of the article in the section Pascal Approximation Method with Optimization in the Stopband, Order 3, LPTF.

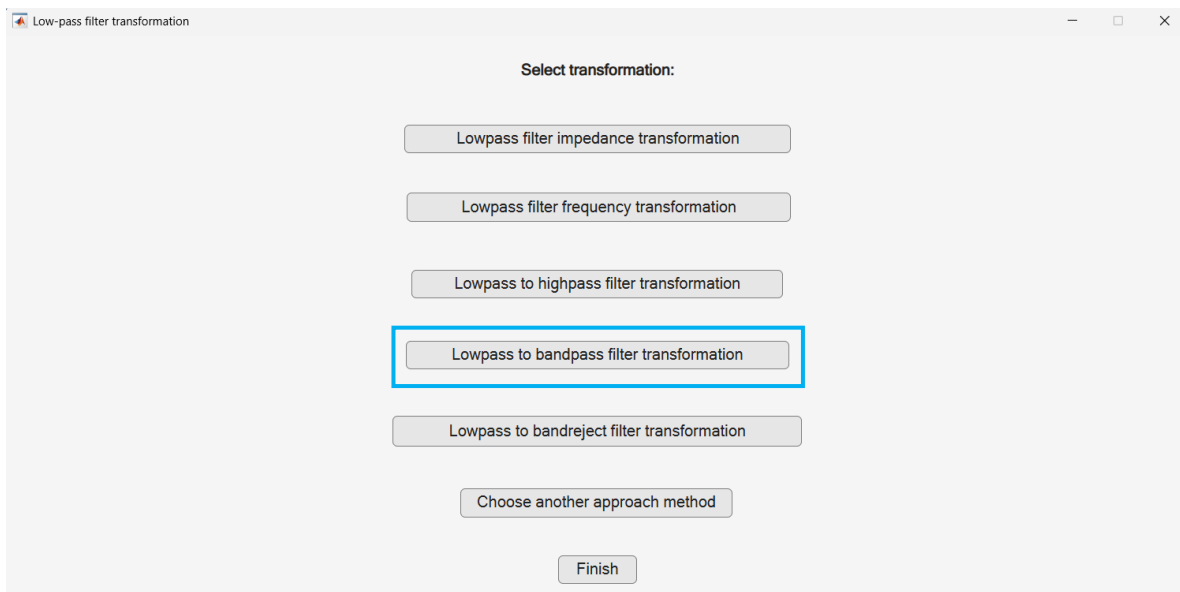
Continuing, the magnitude frequency response is determined.



Then the normalized passive electrical network in π configuration.



The transformation from low-pass filter to band-pass filter is chosen.



The cutoff frequency specifications are entered where it is intended to obtain a result of 50 Hz and 3000 Hz of cutoff frequencies in the frequency response, to obtain these results a tuning is carried out by trial and error in the entry of cutoff frequencies for the transformation. It is found that by entering the frequencies of 55 Hz and 2730 Hz, the cutoff frequencies measured in the frequency response are very close to 50 Hz and 3000 Hz.

Enter Cutoff Frequency 1 [Hz]:
Enter Cutoff Frequency 2 [Hz]:

☒ Optimized Filter in the Stop Band
☐ Optimized Filter in the Pass Band

Transfer Function

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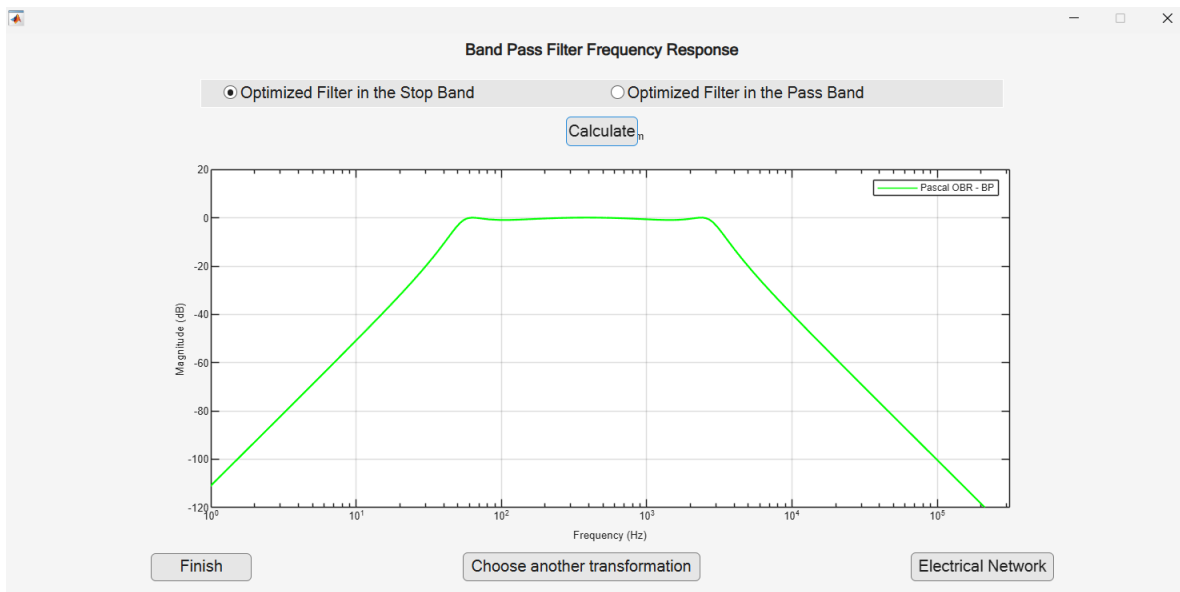
syspmax_bp_f =
2.374e12 s^3
-----
s^6 + 1.661e04 s^5 + 3.676e08 s^4 + 2.53e12 s^3 + 2.179e15 s^2 + 5.837e17 s + 2.083e20
Continuous-time transfer function.
<a href="matlab:disp(char([10 32 32 32 32 32 32 78 117 109 101 114 97 116 111 114 58 32 123 91 48 32 48 32 48 32 50 46 51 55 52 48 101 43 49 50 32 48 32 48 93 125 10 32 32 32 32 68 101 110 111 109 105 110 97 116 111 114 58 32 123 91 49 32 49 46 54 54 49 50 101 43 48 52 32 51 46 54 55 54 50 101 43 48 56 32 50 46 53 50 57 55 101 43 49 50 32 50 46 49 55 57 50 101 43 49 53 32 53 46 56 51 54 57 101 43 49 55 32 50 46 48 56 50 56 101 43 50 48 93 125 10 32 32 32 32 32 32 32 86 97 114 105 97 98 108 101 58 32 39 115 39 10 32 32 32 32 32 32 32 32 32 73 79 68 101 108 97 121 58 32 48 10 32 32 32 32 32 73 110 112 117 116 68 101 108 97 121 58 32 48 10 32 32 32 32 32 79 117 116 112 117 116 68 101 108 97 121 58 32 48 10 32 32 32 32 32 73 110 112 117 116 78 97 109 101 58 32 123 39 39 125 10 32 32 32 32 32 32 73 110 112 117 116 85 110 105 116 58 32 123 39 39 125 10 32 32 32 32 73 110 112 117 116 71 114 111 117 112 58 32 91 49 215 49 32 115 116 114 117 99 116 93 10 32 32 32 32 32 79 117 116 112 117 116 78 97 109 101 58 32 123 39 39 125 10 32 32 32 32 32 79 117 116 112 117 116 85 110 105 116 58 32 123 39 39 125 10 32 32 32 32 79 117 116 112 117 116 71 114 111 117 112 58 32 91 49 215 49 32 115 116 114 117 99 116 93 10 32 32 32 32 32 32 78 111 116 101 115 58 32 91 48 215 49 32 115 116 114 106 110 103 93 10 32 32 32 32 32 32 85 115 101 114 68 97 116 97 58 32 91 93 10 32 32 32 32 32 32 32 32 32 78 97 109 101 58 32 39 39 10 32 32 32 32 32 32 32 32 32 32 84 115 58 32 46 10 32 32 32 32 32 32 32 32 32 32 84 105 109 101 85 110 105 116 58 32 39 115 101 99 111 110 100 115 39 10 32 32 32 83 97 109 112 108 105 110 103 71 114 105 100 58 32 91 49 215 49 32 115 116 114 117 99 116 93 10]])">Model Properties</a>

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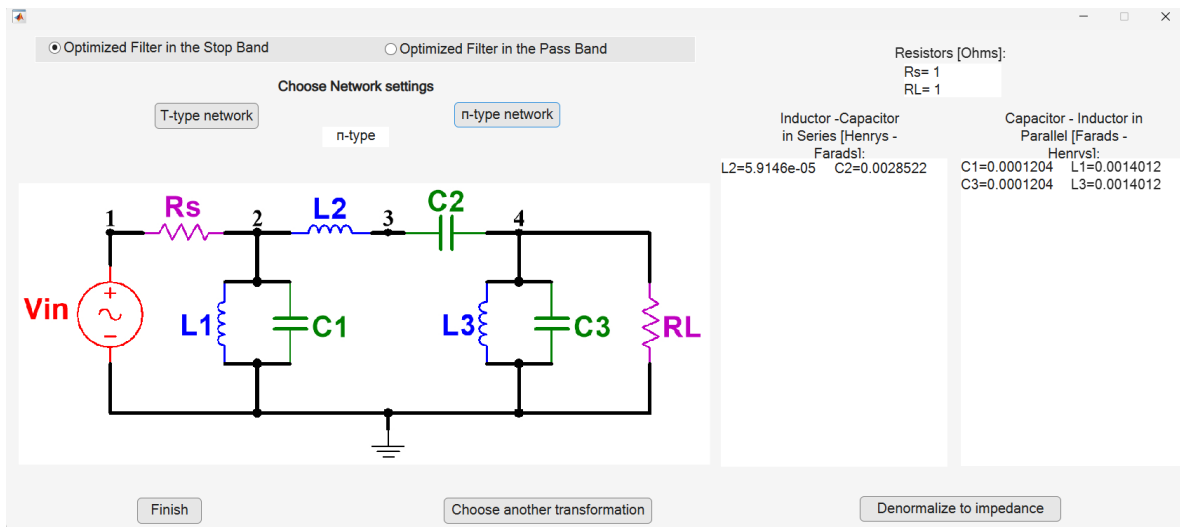
Finish
Frequency response

The transfer function registered in Table III of the article in the section Pascal Approximation Method with Optimization in the Stopband, Order 3, BPTF is obtained.

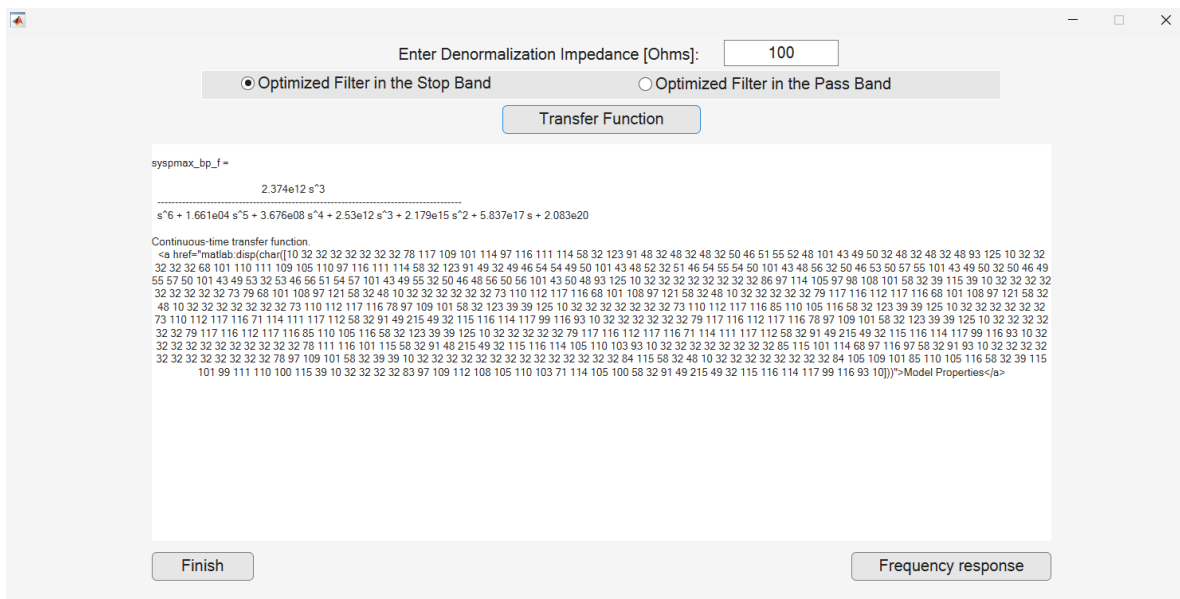
Subsequently the frequency response is generated.



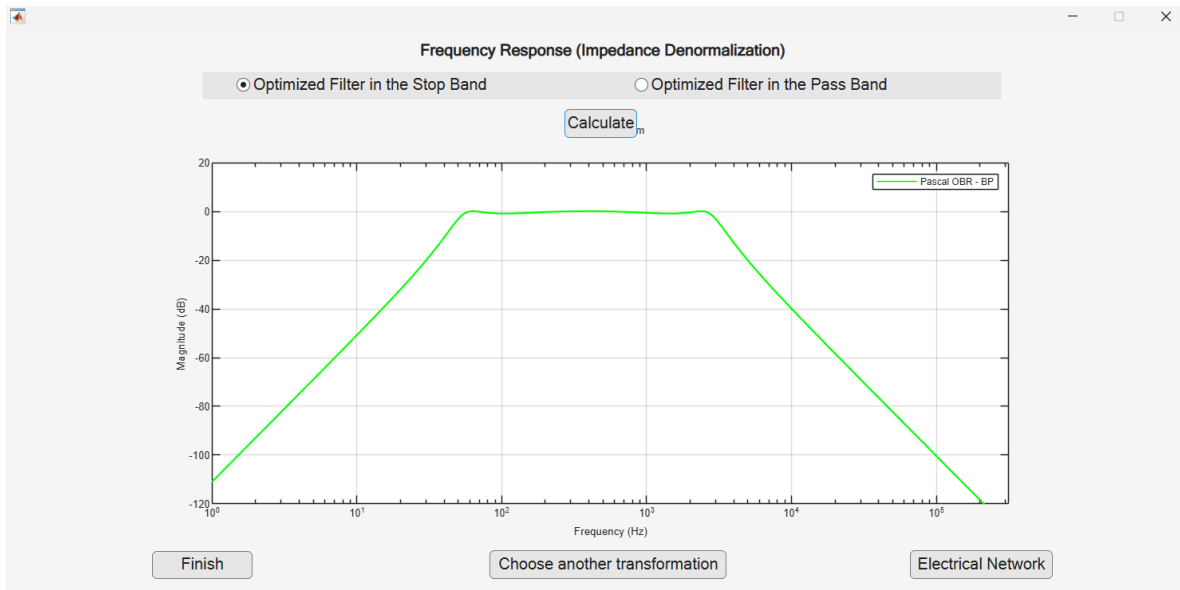
The transformed electrical network is generated.



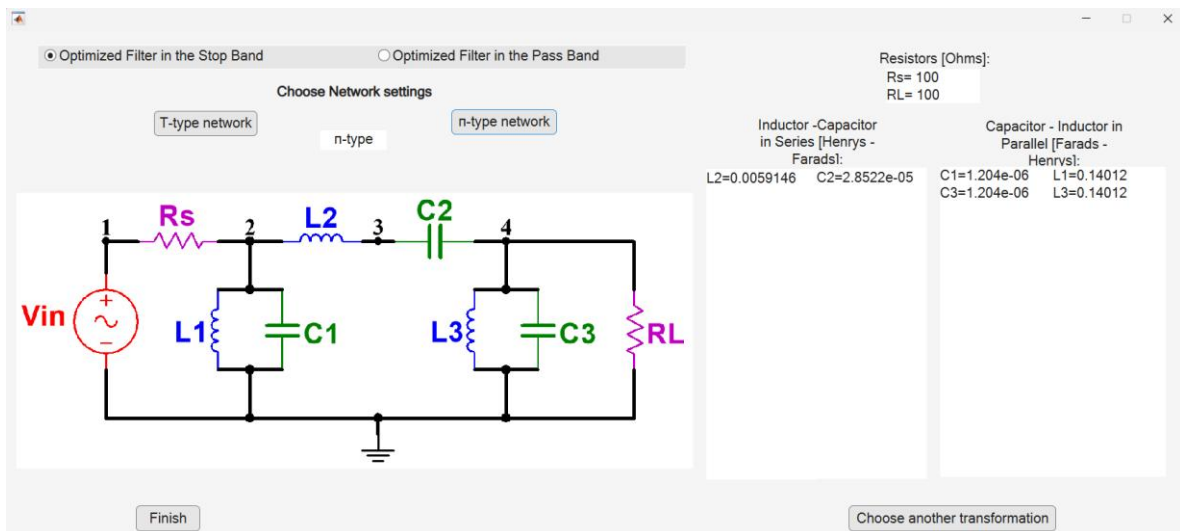
Impedance denormalization is applied with a value of 100Ω .



The transformed frequency response is visualized again.



Finally, the transformed and impedance denormalized electrical network is generated.



The values obtained and the drawing of the electrical network are those shown in Figure 7 of the article, to which the transformation to active elements is externally performed and subsequently it is physically implemented.